

MAGAZINE GALLERY

Deep research report and implementation specification

ISSUE 01	ISSUE 02	ISSUE 03
52 pages	37 pages	74 pages
1224 x 864 px	1224 x 864 px	1224 x 864 px

163

individual page views, each represented by a 153 x 108 px thumbnail and available in an accessible modal viewer.

Strategy / UX / accessibility / SEO / engineering / operations

Prepared 13 July 2026

Executive Summary

The recommended product is a compact, static-first archive: one homepage, one canonical landing page per issue, and a progressively enhanced page viewer. The site should feel editorial, but behave like infrastructure - fast, predictable, linkable, keyboard-operable, and resilient without JavaScript.

The decisive recommendation

Render every thumbnail as a real HTML image inside a real link. JavaScript should intercept that link to open a native modal dialog, while the link still reaches the full page image when scripting is unavailable. Load only thumbnails on the gallery route; fetch the 1224 x 864 page image on demand and cautiously prefetch adjacent pages after the viewer opens.

Decision	Recommendation	Reason
Delivery	Pre-rendered HTML on an edge CDN	Only four primary content pages; excellent cacheability and low operational overhead.
Gallery	Complete DOM, native lazy loading	All 163 pages remain crawlable and keyboard reachable without client rendering.
Viewer	Native <dialog> + progressive enhancement	Built-in top-layer, Escape, inert background, and focus behavior; still test with assistive tech.
Images	AVIF/WebP with JPEG fallback	Modern compression plus robust compatibility and crawlable .
Conformance	WCAG 2.2 AA release gate	Explicit keyboard, contrast, reflow, target-size, motion, and focus acceptance tests.

Success measures

- All 163 expected thumbnails present, uniquely numbered, correctly sized, and linked to a valid full page.
- Core Web Vitals at the 75th percentile: LCP <= 2.5 s, INP <= 200 ms, CLS <= 0.1 on mobile and desktop [6].
- Zero critical or serious automated accessibility findings, plus keyboard, screen-reader, zoom, and reflow verification.
- Issue pages discoverable through navigation, canonical URLs, XML sitemap, and image sitemap; no modal-state duplicate indexing.

Scope, Inventory, and Assumptions

Issue	Pages	Master image	Thumbnail	Scale	Desktop rows @ 7 cols
Issue 01	52	1224 x 864	153 x 108	12.5% each axis	8
Issue 02	37	1224 x 864	153 x 108	12.5% each axis	6
Issue 03	74	1224 x 864	153 x 108	12.5% each axis	11
Total	163	199,512,576 pixels	2,693,412 pixels	Exact 8:1 reduction	25 rows

The master and thumbnail ratios are both 17:12 (approximately 1.4167:1). The requested thumbnail is therefore a mathematically exact one-eighth linear derivative: $1224 / 8 = 153$ and $864 / 8 = 108$. No crop, letterbox, or focal-point logic is needed.

Asset arithmetic

Set	Count	Purpose
Full-page fallback images	163	Direct links and modal source fallback
Thumbnail fallback images	163	Gallery src fallback
Core fallback assets	326	Minimum raster inventory
If AVIF + WebP + JPEG are produced	978	Three encodings for each of 326 logical derivatives

Assumptions: page images are already color-correct and rights-cleared; OCR and searchable full text are out of scope unless separately supplied; downloads are optional; issues have editorial titles/dates even though placeholders are used here. The report treats 153 x 108 as the thumbnail asset dimension, not an immutable CSS box on very narrow screens.

Risk register

Risk	Effect	Control
Every full page loads on route entry	Multi-megabyte page; slow mobile experience	Lazy-load thumbnails; request full page only in viewer.
Page image is the only content	Poor search/a11y context	Issue-level title, summary, date, captions, and page labels in HTML.
Modal has custom focus code	Keyboard trap or focus loss	Prefer native dialog; regression-test Tab, Shift+Tab, Escape, and focus return.
Numbering drift	Missing or duplicated page	Manifest validator enforces contiguous 001..N and exact dimensions.

Functional Requirements

ID	Priority	Requirement / acceptance criterion
FR-01	Must	Homepage presents all three issues with cover, title, date/season, page count, short synopsis, and issue link.
FR-02	Must	Each issue route renders exactly one ordered thumbnail for every page: 52, 37, and 74 respectively.
FR-03	Must	Activating any thumbnail opens its corresponding 1224 x 864 page in the viewer without reloading the issue page.
FR-04	Must	Previous/next controls, ArrowLeft/ArrowRight, and page indicator move through the issue; movement stops or wraps according to a documented policy. Recommended: stop at ends.
FR-05	Must	Escape and a visible Close control dismiss the viewer; focus returns to the originating thumbnail link.
FR-06	Must	Browser Back closes a deep-linked viewer state before leaving the issue route. Use a page fragment such as #page-017 only as UI state, not as separate indexable content.
FR-07	Must	With JavaScript disabled, thumbnail links open the full fallback image and issue navigation remains complete.
FR-08	Must	The current page, issue title, and position (for example, Page 17 of 52) are available visually and to assistive technology.
FR-09	Should	Zoom supports fit, 100%, 150%, and 200%; panning never blocks the Close or navigation controls.
FR-10	Should	After viewer open, prefetch only previous and next full pages when Save-Data is off and the connection is not constrained.
FR-11	Should	Analytics records issue_view, page_viewer_open, page_navigate, zoom_change, and viewer_close without collecting image content or sensitive data.

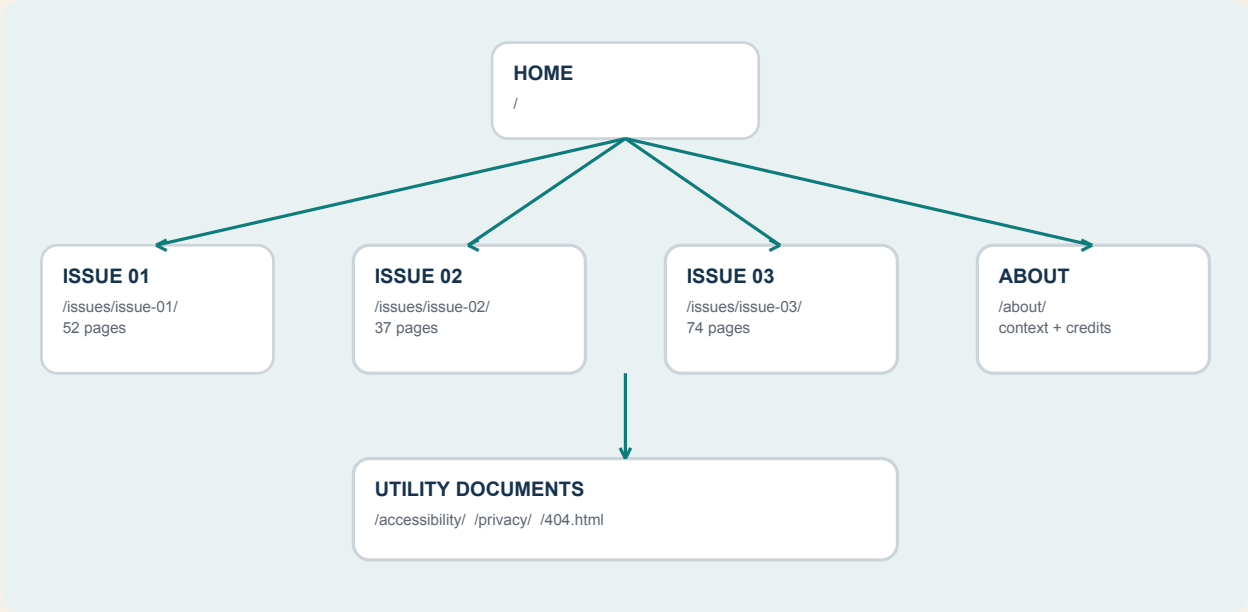
Non-functional release gates

- Compatibility: latest two major versions of Chromium, Firefox, and Safari; current iOS Safari and Android Chrome; graceful fallback elsewhere.
- Performance: initial issue page transfers thumbnails only; no eagerly loaded full-resolution page except a deliberate above-the-fold editorial cover.
- Reliability: immutable, content-hashed image URLs; atomic deployment; previous release retained for rollback.
- Privacy: no third-party viewer dependency; analytics delayed until consent where required; honor policy and regional obligations.

04 FINDABILITY WITHOUT COMPLEXITY

Information Architecture and Sitemap

A shallow hierarchy is sufficient. The homepage is the editorial front door; issue pages are durable archive nodes. Individual modal states are interface states, not separate crawl targets, unless the publisher later adds page-level transcriptions worth indexing.

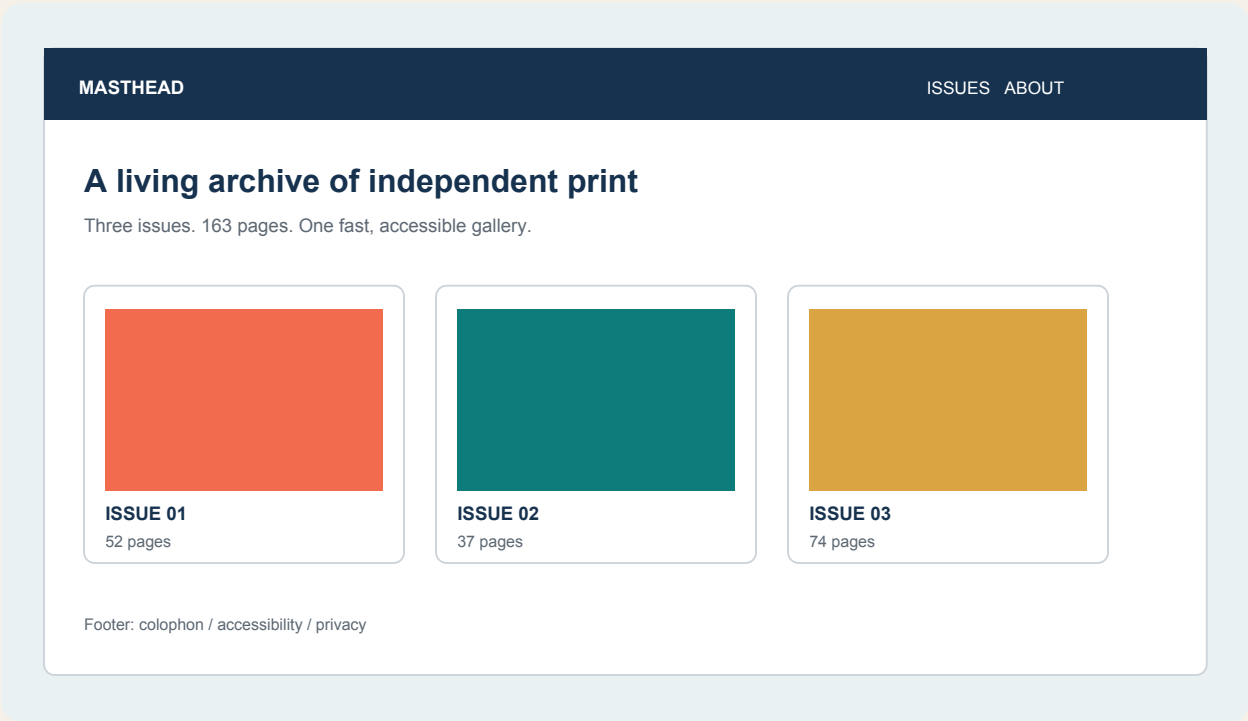


Navigation model

Layer	Desktop	Mobile	Notes
Global	Masthead, Issues, About	Masthead + Menu button	Keep issue access within one activation.
Issue-local	Issue title, page count, jump-to-page	Compact title + jump control	Sticky controls may be used only if focused content is not obscured [2].
Viewer	Close, previous, next, zoom, position	Same controls with larger touch targets	Do not hide essential functions behind hover.
Footer	Accessibility, privacy, colophon	Same links stacked	Include a plain contact path for accessibility feedback.

Recommended canonical routes: `/`, `/issues/issue-01/`, `/issues/issue-02/`, `/issues/issue-03/`, and `/about/`. Add `/accessibility/` and `/privacy/` as governance pages. Keep page-image asset URLs stable, descriptive, and consistent across all references.

Homepage Design



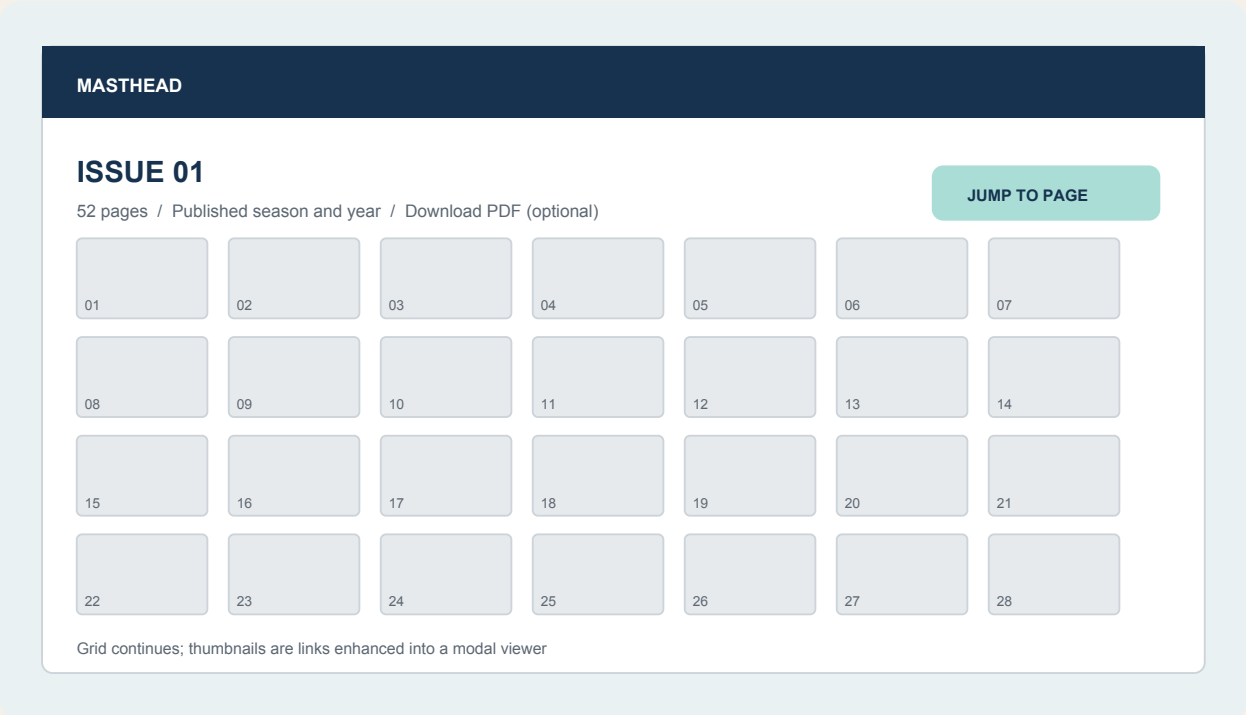
Content hierarchy

- Masthead and one-sentence archive proposition establish purpose before decoration.
- Three issue cards are the primary actions. Each card uses the issue cover as content, not as a CSS background, so it remains crawlable [8].
- Card metadata follows a consistent order: issue number/title, date or season, page count, synopsis, and View issue link.
- A compact colophon can explain publisher, archive scope, and image rights. Avoid a carousel: all three issues fit on one responsive page and should remain simultaneously discoverable.

Homepage acceptance criteria

Area	Specification
Hero	Text-first; optional cover montage below or beside copy. If a hero image is LCP, do not lazy-load it and consider fetchpriority=high [7].
Issue cards	3 across >= 960 px; 2 across at medium widths; 1 column on narrow screens. Entire visual may be linked, but preserve a clear accessible name.
Typography	Editorial display face may be used for headings; UI/body stays highly legible. Cap body measure near 65 characters.
Motion	Subtle opacity/transform only; remove or greatly reduce under prefers-reduced-motion.

Magazine Issue Pages



The issue page combines editorial context with a complete ordered gallery. Page numbers belong beneath or overlaid within each thumbnail only if contrast remains robust; the link's accessible name should be explicit: 'Open Issue 01, page 17 of 52'.

Recommended page anatomy

- Breadcrumb: Home / Issue 01. Use semantic navigation and aria-current on the issue.
- H1 with issue title; metadata line; short synopsis; optional credits and downloadable whole-issue PDF.
- Jump-to-page control implemented as a labeled select or numeric input plus Go, not a custom combobox unless needed.
- Ordered list of page links. DOM order equals reading and visual order. Do not virtualize 74 small thumbnails: the complexity and accessibility cost outweigh the savings.
- Previous/next issue links after the gallery, followed by the global footer.

Page-count integrity

Issue	First valid	Last valid	Expected links	Build assertion
01	001	052	52	set(manifest pages) == set(1..52)
02	001	037	37	set(manifest pages) == set(1..37)
03	001	074	74	set(manifest pages) == set(1..74)

Gallery Layouts and Thumbnail Grid

At wide desktop sizes, seven native-size thumbnails fit comfortably in a 1200 px content region: 7 x 153 + 6 x 16 = 1167 px, leaving 33 px for centering. The grid should use CSS Grid and explicit gaps, not percentage arithmetic or JavaScript measurement.

Viewport / container	Columns	Cell target	Gap	Rows: I1 / I2 / I3
>= 1180 px content	7	153 x 108 px	16 px	8 / 6 / 11
720-1179 px content	4 or 5	153 x 108 px	16 px	13 / 10 / 19 (4 cols)
540-719 px content	3	153 x 108 px	14 px	18 / 13 / 25
<= 539 px content	2	minmax(132,153) x auto	12 px	26 / 19 / 37

Recommended CSS contract

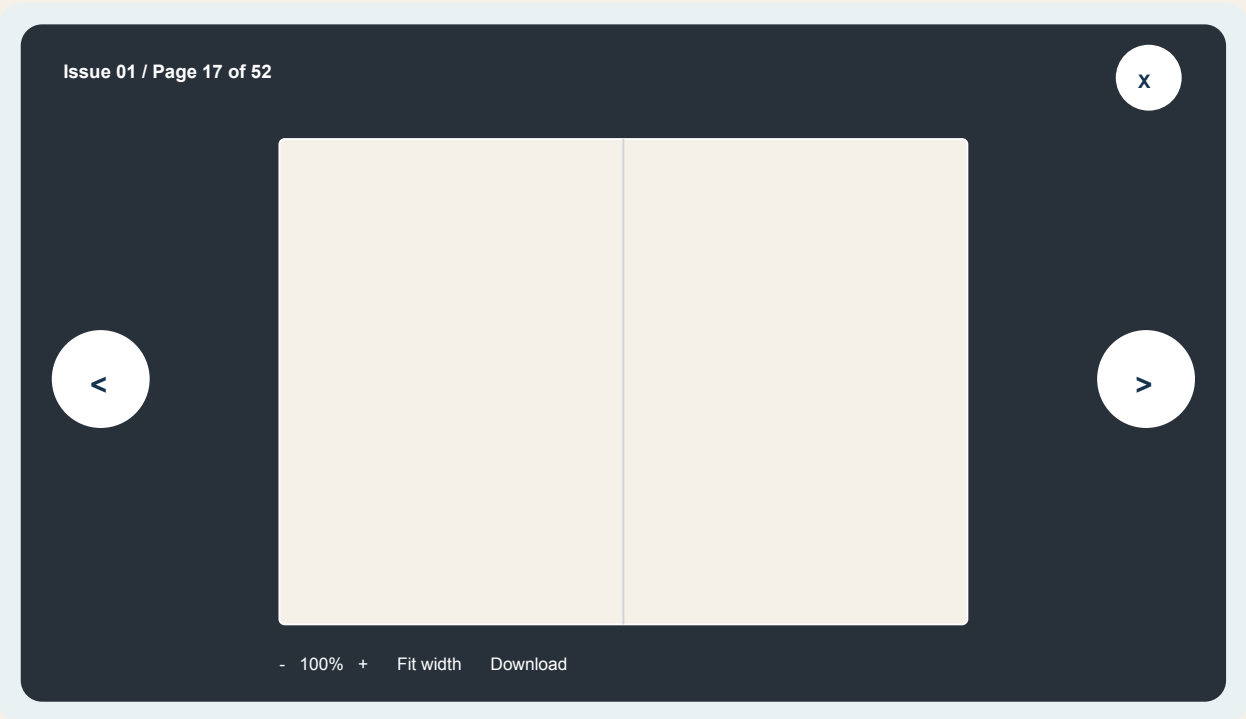
```
.page-grid {
display: grid;
grid-template-columns: repeat(auto-fit, minmax(min(132px, 100%), 153px));
justify-content: center;
gap: clamp(12px, 1.5vw, 16px);
list-style: none; margin: 0; padding: 0;
}
.page-thumb { width: 100%; aspect-ratio: 17 / 12; object-fit: contain; }
```

Use the intrinsic width=153 and height=108 attributes even if CSS scales the image. These attributes reserve the correct aspect ratio and reduce layout shift [9]. Each link's interactive area should include the whole image and label; target at least 24 x 24 CSS px under WCAG 2.2 AA, while a 44 x 44 design target is preferable for prominent controls [3].

Grid visual rules

- Neutral matte behind each thumbnail so white page edges remain visible; 1 px boundary with at least 3:1 non-text contrast where needed.
- No justified-masonry layout: page order must remain obvious and rows aligned.
- Visible page number outside the raster. Do not burn UI labels into the image.
- Hover may lift by 2 px; focus uses a persistent 2-3 px two-color outline that survives mixed imagery.

Modal Viewer Specifications



Interaction contract

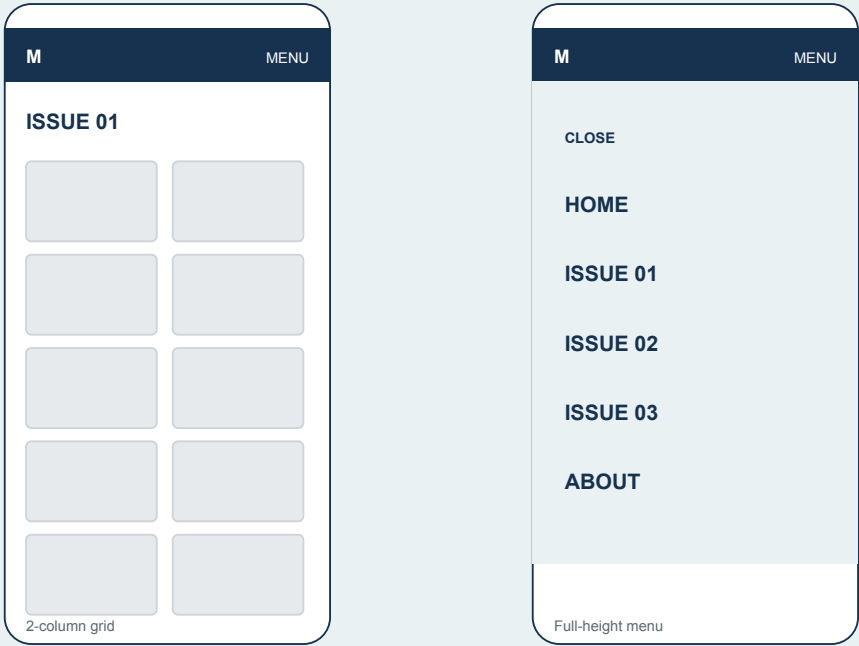
Input	Behavior
Thumbnail activation	Save invoker; set image and labels; call <code>dialog.showModal()</code> ; focus Close or static title depending on viewport/reading context.
Escape / Close	Close viewer, restore scroll position, return focus to invoker.
ArrowLeft / ArrowRight	Navigate when focus is not in an editable control. Announce new page position in a polite live region.
Tab / Shift+Tab	Remain within the modal. Native modal dialog supplies inert background; verify cycling in target browsers [4][5].
Backdrop click	Optional light dismiss. Never make it the only close method.
Swipe	Optional enhancement; buttons and keyboard remain primary equivalents.

The native `<dialog>` element called with `showModal()` provides top-layer rendering, modal semantics, inert background, and Escape handling. It has broad cross-browser availability, but implementation still needs assistive-technology testing [5]. The dialog must have a visible title or accessible label. Do not put `tabindex` on `<dialog>` itself.

Viewport and zoom

- At fit mode, constrain image to `min(1224 px, available width)` and `min(864 px, available height after controls)`. Preserve 17:12.
- At zoom > 100%, pan only the image stage; keep Close, navigation, and page position fixed and reachable.
- Use overscroll-behavior: contain on the stage and lock background scrolling without resetting the underlying scroll position.
- Expose the same caption and page position regardless of zoom. Never rely on pinch alone.

Mobile Navigation and Responsive Layouts



Breakpoint strategy

Choose breakpoints where content needs them, not by device brand. The core transitions are navigation collapse, issue-card stacking, thumbnail column changes, and modal controls moving to a bottom toolbar.

Range	Navigation	Gallery	Viewer
>= 960 px	Inline global links	5-7 columns	Side previous/next; bottom zoom bar
600-959 px	Inline or compact menu	3-4 columns	Controls at edges; image fitted
320-599 px	Menu button opens modal/off-canvas nav	2 columns, fluid cells	Bottom controls, 44 px design targets
At 320 CSS px	No horizontal page scroll	Cells may render below 153 px; source remains 153 x 108	Image stage scrolls only when zoomed

- Use a real button with expanded state and an accessible label for the menu. Keep Home, all three issues, About, Accessibility, and Close reachable by keyboard.
- The menu should not cover a focused control behind it. A modal menu is simplest: inert the background, trap focus, support Escape, and restore focus.
- Support portrait and landscape; safe-area insets; 200% browser zoom; 400% text zoom/reflow at 320 CSS px equivalent.
- Avoid sticky chrome that consumes the limited viewer height. If controls auto-hide, they must reappear on focus, pointer movement, or tap and never hide the focused element.

Accessibility: WCAG 2.2 AA

WCAG 2.2 became a W3C Recommendation on 5 October 2023 and adds nine success criteria over 2.1 [1]. For this interface, the highest-risk areas are image alternatives, keyboard access, focus visibility/restoration, modal semantics, target size, reflow, contrast, and status announcements.

Criterion	Level	Implementation / test
1.1.1 Non-text Content	A	Every page image has context-appropriate alt. If no transcription exists: 'Issue 01, page 017' plus concise page subject where editorially known.
1.3.1 Info and Relationships	A	Landmarks, H1/H2 hierarchy, ordered list for pages, native buttons and links.
1.4.3 Contrast	AA	Normal text $\geq 4.5:1$; large text $\geq 3:1$. Verify every state, not only default.
1.4.10 Reflow	AA	No two-dimensional page scroll at 320 CSS px except the intentionally zoomed page canvas.
1.4.11 Non-text Contrast	AA	Controls, selected states, and focus boundaries $\geq 3:1$ against adjacent colors.
2.1.1 Keyboard	A	Every gallery and viewer action works with keyboard; no keyboard trap.
2.4.3 / 2.4.7 Focus	A/AA	Logical DOM order; strong visible focus; focus returns from viewer.
2.4.11 Focus Not Obscured	AA	Sticky headers/toolbars never fully cover focused items [2].
2.5.8 Target Size (Minimum)	AA	At least 24 x 24 CSS px or permitted spacing; design primary controls at 44 x 44 [3].
4.1.2 Name, Role, Value	A	Dialog, buttons, expanded menu, zoom state, and page controls expose correct programmatic state.

Accessibility Test Matrix

Test	Method	Pass condition
Keyboard gallery	Tab from skip link through issue page	Order follows pages 001..N; every link visible on focus; Enter opens correct page.
Dialog keyboard	Open middle page; cycle Tab/Shift+Tab; use arrows; Escape	Focus remains inside, navigation works, Escape closes, invoker regains focus [4].
Screen reader	VoiceOver/Safari and NVDA/Firefox	Dialog name and Page X of N announced once; image alternative useful; state changes not noisy.
Zoom/reflow	200% zoom and 400% text; 320 CSS px equivalent	No clipped navigation or page-level horizontal scroll; controls remain operable.
Contrast	Automated scan plus manual state sampling	Text, UI, hover, focus, disabled, backdrop and image-adjacent controls pass.
Touch	iOS and Android, portrait/landscape	No accidental adjacent activation; controls remain clear near safe areas.
Reduced motion	OS reduce motion enabled	Viewer and menu transitions are instant or subtle; no essential information depends on animation.
Images off / failures	Disable images; force 404	Alt/context remains; broken full image produces recoverable message and controls stay usable.

Alt-text policy

A scanned magazine page is complex content. A generic number-only alt is adequate for navigation only when the adjacent page label already supplies the same information and a longer description would not convey the page's full content. If editorial metadata exists, prefer a short subject description and provide a transcription or structured summary nearby. Do not paste OCR into alt attributes; long text belongs in expandable page notes or a dedicated transcription.

Release evidence

- Automated reports from axe-core or equivalent for all four templates/states.
- Keyboard test recording or checklist for first, middle, and last page in each issue.
- Screen-reader notes including browser/AT/version and known limitations.
- Contrast calculations for the approved palette and focus treatment.
- Accessibility statement with conformance target, known exceptions, and contact route.

Image SEO and Content Discoverability

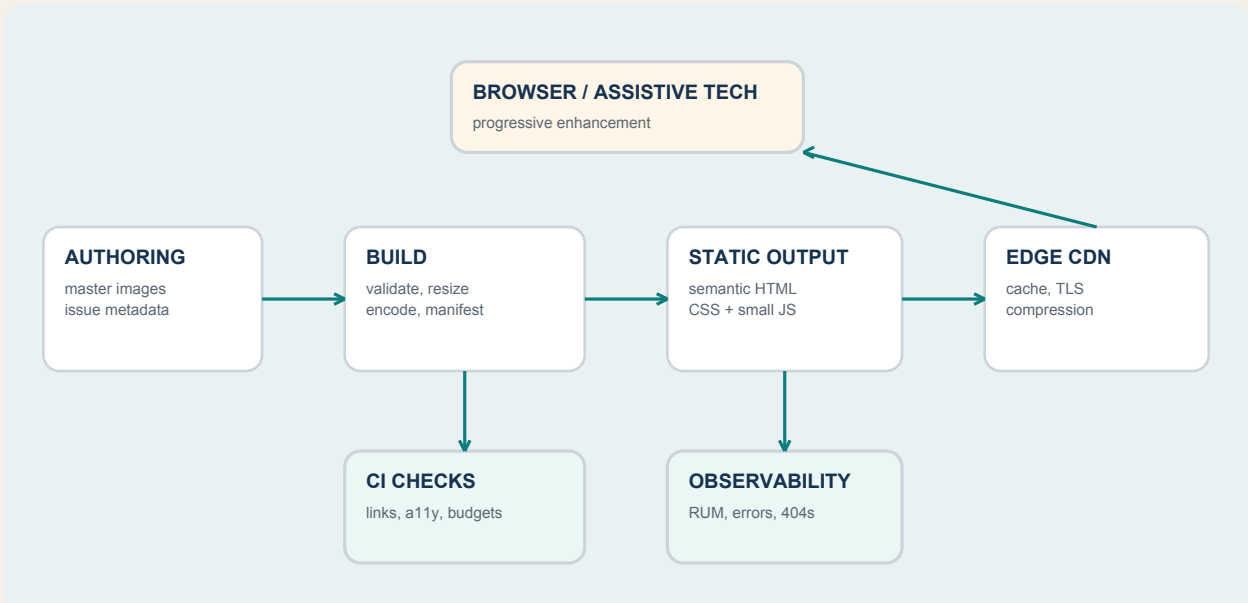
Google recommends standard `<img>` elements, descriptive filenames and alt text, responsive sources with a `src` fallback, relevant surrounding copy, high-quality images, and image sitemaps where discovery matters [8]. Google can discover `<img src>` resources, including inside `<picture>`, but does not index CSS background images [8].

Layer	Recommendation
Landing page	Unique title, meta description, H1, synopsis, date, issue number, publisher/creator credits, and self-referencing canonical.
Image markup	Use <code><picture></code> with AVIF/WebP sources and an <code></code> fallback; provide width/height, alt, decoding, loading, <code>srcset</code> , and sizes.
Names	issue-01-page-017-thumb-153x108.webp and issue-01-page-017-full-1224x864.webp; avoid image17.jpg.
Structured data	CreativeWork or PublicationIssue where applicable, with representative cover image; validate syntax and eligibility. Do not promise a rich result.
Sitemaps	Root XML sitemap for canonical HTML routes plus image entries or an image sitemap. A sitemap helps discovery but does not guarantee indexing [10].
Social previews	Issue cover as og:image with width, height, and meaningful alt; use a high-resolution, representative image [8].
URLs	Stable lowercase paths; HTTPS; no session IDs. Keep modal fragments out of sitemap and canonicalize to the issue route.

Indexing decision

Recommended now: index the homepage and three issue pages, not 163 thin page routes. Reconsider only if each page gains meaningful, unique text such as article title, authorship, abstract, transcript, or archival notes. Search engines can usually discover a well-linked small site, but this image-heavy archive still benefits from a sitemap [10].

Technical Architecture



For four canonical content routes and immutable image assets, a server-rendered application is unnecessary. A small static-site build can convert issue manifests into semantic HTML, validate the inventory, and publish to any CDN-backed object host. This reduces runtime failure modes and makes rollback a file-level operation.

Recommended components

Component	Responsibility	Constraint
Content manifests	Issue title, slug, date, page count, per-page metadata	Schema-validated and reviewed in version control
Build generator	Create pages, picture markup, sitemaps, integrity report	Deterministic output; no production database
Image pipeline	Validate masters; derive and encode assets	Never upscale; strip unsafe metadata; preserve desired color profile
Client JS	Viewer state, navigation, zoom, menu	Target <= 20 KB minified + compressed for first-party code
CDN/host	TLS, compression, caching, redirects, security headers	Atomic deploy and immutable asset caching
Monitoring	RUM, JS errors, broken assets, synthetic checks	Privacy-aware, sampled, alert on regressions

HTML Structure

```
<main id="main">
<nav aria-label="Breadcrumb">...</nav>
<header><h1>Issue 01</h1><p>52 pages</p></header>
<ol class="page-grid">
<li id="page-017">
<a href="/assets/issues/issue-01/pages/issue-01-page-017-full-1224x864.jpg"
data-page="17" aria-label="Open Issue 01, page 17 of 52">
<picture>
<source type="image/avif" srcset="...avif">
<source type="image/webp" srcset="...webp">

</picture><span>Page 17</span>
</a>
</li>
</ol>
</main>
<dialog id="viewer" aria-labelledby="viewer-title">...</dialog>
```

The image alt is empty in this pattern because the link already has a complete accessible name and the adjacent visible label provides the page number; repeating both can be noisy. If the page image has a useful subject description, put it in the link label or visible caption and adjust the alt strategy after screen-reader testing.

Progressive enhancement rules

- Keep a usable href. JavaScript may preventDefault only after dialog support and required viewer elements are confirmed.
- Do not populate the gallery exclusively from JSON in the browser. HTML must contain all links at response time.
- The first visible thumbnail may use loading=eager; later thumbnails use loading=lazy. Never lazy-load a likely LCP hero image [7].
- Use a live region for page-position updates, but avoid announcing every incidental zoom or pan movement.

CSS Architecture

Layer	Examples	Rule
Settings	color, type, spacing, width, z-index, motion tokens	Custom properties in :root; tested color pairs
Base	box sizing, body, images, headings, links	Low specificity; no blanket outline removal
Objects	container, flow, cluster, grid	Layout only; reusable and content-agnostic
Components	masthead, issue-card, page-grid, viewer, menu	One component owns its states and parts
Utilities	visually-hidden, no-scroll, stack overrides	Single purpose and documented
States	[open], [aria-current], [data-zoom], :focus-visible	Prefer attributes over extra state classes

Core tokens

```
:root {
  --content: 75rem; --thumb-w: 153px; --thumb-ratio: 17 / 12;
  --space-1: .25rem; --space-2: .5rem; --space-3: .75rem;
  --space-4: 1rem; --space-6: 1.5rem; --space-8: 2rem;
  --focus-inner: #fff; --focus-outer: #073b4c;
}
:focus-visible { outline: 3px solid var(--focus-outer); outline-offset: 3px; }
@media (prefers-reduced-motion: reduce) { *, *::before, *::after { scroll-behavior: auto !important; animation-duration: .01ms !important; } }
```

A single stylesheet should be adequate. Container queries can tune issue-card internals, while a small number of viewport media queries control global navigation and viewer chrome. Keep critical styles small and avoid CSS-in-JS for this static archive.

JavaScript Architecture

Module	Responsibility	Public events / state
viewer-controller	Open/close, current page, history fragment, focus return	closed opening open closing
viewer-image	Source selection, decode, error/retry, adjacent prefetch	idle loading ready error
viewer-controls	Previous/next, zoom, keyboard routing, live status	pageIndex, zoomMode, zoomLevel
mobile-menu	Expanded state, modal behavior, focus restore	open boolean
analytics-adapter	Consent-aware custom events	No-op if unavailable or denied

Behavioral sequence

- Delegate click handling from the ordered list; resolve the clicked link and manifest record.
- Store the invoker. Update title, count, image sources, caption, prev/next disabled states, and #page-NNN.
- Open native dialog. Wait for image.decode() when supported; replace skeleton with image; expose a recoverable error on failure.
- On navigation, update the same dialog in place; abort stale fetch/decode work; announce Page X of N.
- On close, restore the prior fragment/history state, unlock scroll, and focus the stored link without scrolling unexpectedly.

Guardrails

- No dependency is required. If one is introduced, justify its transfer and maintenance cost.
- Do not bind 163 individual listeners; use event delegation.
- Avoid inserting unsanitized manifest strings with innerHTML. Use textContent and attribute assignment.
- Use AbortController for superseded image work; handle rapid arrow-key input without showing stale pages.
- Respect navigator.connection.saveData where available; prefetch is an optimization, never correctness-critical.

Image Optimization and Performance

The issue page should transfer only the images needed for orientation and the upcoming viewport. Browser-level `loading=lazy` provides native deferred loading without a custom library [7]. Responsive sources and accurate sizes let the browser choose an appropriate resource [9].

Asset	Preferred delivery	Loading policy	Quality target
Issue cover / possible LCP	AVIF/WebP/JPEG picture	Eager; <code>fetchpriority=high</code> only when it is the known LCP image	Visually lossless at display size
153 x 108 thumbnails	AVIF/WebP/JPEG; intrinsic dimensions	First row eager or auto; remaining lazy; <code>decoding=async</code>	Text still recognizable; no ringing around type
1224 x 864 modal pages	AVIF/WebP/JPEG; optional 612w + 1224w srcset	On viewer demand; adjacent prefetch after open	Editorial text readable at 100%; preserve line art

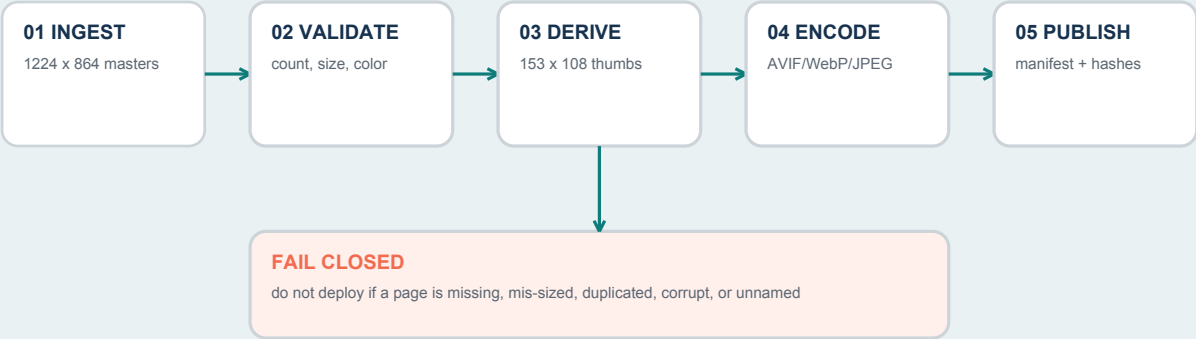
Budgets and headers

- Set Cache-Control: public, max-age=31536000, immutable on hashed images; short revalidation on HTML.
- Compress HTML/CSS/JS with Brotli or gzip. Image formats are already compressed and should not be recompressed by the server.
- Reserve image geometry with width/height and CSS aspect-ratio to protect CLS [9].
- Performance budget: ≤ 200 KB compressed HTML+CSS+JS for an issue page excluding images; ≤ 350 KB images above the fold on a representative mobile viewport; no initial full-page viewer image.
- Measure field performance at the 75th percentile separately for mobile and desktop. Good thresholds are LCP ≤ 2.5 s, INP ≤ 200 ms, CLS ≤ 0.1 [6].

Quality testing

Magazine pages combine photographs, small typography, flat color, and line art; a single encoder quality may fail unpredictably. Establish a representative corpus (cover, text-heavy page, dark spread, fine line art, photo page), compare AVIF/WebP/JPEG at 100% and 200%, and set objective ceilings only after visual approval.

Thumbnail Generation Pipeline



Stage	Checks / output
Ingest	Read manifest; verify master is decodable, RGB/sRGB-compatible, exactly 1224 x 864, and mapped to one issue/page.
Validate	Expected counts 52/37/74; contiguous numbering; no duplicate checksums unless intentionally documented; no unexpected alpha or animation.
Derive	Resize to exactly 153 x 108 with a high-quality downsampling filter. Do not crop or sharpen aggressively. Optionally apply mild output sharpening after review.
Encode	Create AVIF and WebP plus JPEG fallback for full and thumbnail logical assets. Strip EXIF/GPS; keep required rights metadata in HTML/manifest.
Verify	Decode outputs; assert dimensions, format, byte ceiling, perceptual-quality threshold or approved fixture comparison; produce contact sheet.
Publish	Write deterministic paths; compute hashes; generate HTML/picture markup, sitemaps, and asset inventory; upload atomically.

Pipeline invariants

- One master produces one 153 x 108 thumbnail without crop: the ratio is identical.
- A build with 162 or 164 pages fails. A build with page 016 twice and no 017 fails.
- Source masters are never overwritten; derivatives live in generated directories.
- The same input bytes plus tool versions and configuration produce the same filenames and manifest.
- CI archives an integrity report: counts, dimensions, bytes, checksums, and encoder versions.

File Naming and Folder Structure

Naming convention

Use lowercase ASCII, hyphens, zero-padded three-digit page numbers, explicit role, and explicit dimensions. Human-readable names give search engines light context and make operational errors easier to spot [8].

Purpose	Pattern	Example
Master	issue-{NN}-page-{NNN}-master.{ext}	issue-01-page-017-master.tif
Full fallback	issue-{NN}-page-{NNN}-full-1224x864.jpg	issue-01-page-017-full-1224x864.jpg
Full modern	...full-1224x864.{webp avif}	issue-01-page-017-full-1224x864.avif
Thumbnail	...thumb-153x108.{jpg webp avif}	issue-01-page-017-thumb-153x108.webp
Cover	issue-{NN}-cover-{width}x{height}.{ext}	issue-02-cover-1224x864.webp

Recommended tree

```
project/  
content/issues/  
issue-01.json  
issue-02.json  
issue-03.json  
source/issues/issue-01/masters/  
src/  
templates/ styles/ scripts/  
public/  
assets/issues/issue-01/  
pages/full/ pages/thumb/  
sitemap.xml image-sitemap.xml robots.txt  
tools/images/  
tests/  
accessibility/ visual/ integrity/  
dist/
```

If content hashing is supported, append a short hash before the extension and keep the descriptive stem, for example issue-01-page-017-thumb-153x108.a1b2c3.webp. Never encode editorial title text that may change into a permanent path unless redirects are guaranteed.

Deployment, Security, and Operations

Phase	Actions	Gate
Pull request	Build site; validate manifests/images; lint; unit tests; link check; generate preview	All expected assets and routes present
Preview QA	Responsive visual review; keyboard; screen reader sample; image contact sheet; Lighthouse/axe	Signed checklist; no severe defects
Production publish	Upload hashed assets first, then HTML; atomic alias/symlink switch	Smoke checks pass from multiple regions
Post-release	Submit sitemap; monitor 404s, RUM, JS errors, CDN hit rate	No regression over baseline
Rollback	Repoint to prior immutable release; retain image assets referenced by cached HTML	Recovery objective measured in minutes

Headers

- Content-Security-Policy: default-src 'self'; img-src 'self' data: approved-CDN; script-src 'self'; style-src 'self'; object-src 'none'; base-uri 'self'; frame-ancestors 'none'. Tune in report-only mode first.
- Strict-Transport-Security after HTTPS is stable; X-Content-Type-Options: nosniff; Referrer-Policy: strict-origin-when-cross-origin; Permissions-Policy disabling unused capabilities.
- HTML: Cache-Control with short max-age and revalidation. Hashed assets: one-year immutable. 404 page: noindex and useful issue links.
- If the asset CDN uses another hostname, verify it in Search Console and include absolute image URLs in the image sitemap [8].

Observability

Monitor route availability, broken image responses, asset latency, viewer errors, issue/page mismatch events, and Core Web Vitals. Keep dashboards segmented by issue and viewport class. Alert on sustained failures, not individual user connectivity errors. Review privacy implications before recording page-level interaction analytics.

Quality Assurance and Acceptance Plan

Domain	Automated	Manual	Release threshold
Inventory	Manifest schema, count, contiguous IDs, dimensions, checksums	Contact-sheet scan	52/37/74 exact; zero missing/duplicate
Functional	Viewer state and boundary unit tests; link checker	Mouse, touch, keyboard, Back, no-JS	All FR Must items pass
Responsive	Screenshot matrix at selected widths	Real iOS/Android portrait + landscape	No clipping or unintended horizontal page scroll
Accessibility	axe or equivalent; HTML validation	NVDA/Firefox, VoiceOver/Safari, zoom/reflow	Zero critical/serious; checklist signed
Performance	Lighthouse CI budgets, image byte checks	Throttled viewer experience	Budgets pass; no initial full image
SEO	Canonical/sitemap/schema/link validation	Rendered-source and Search Console inspection	Canonical HTML and images discoverable
Security	Header checks, dependency scan	CSP report review	No high-severity unresolved issue

Representative viewport matrix

Viewport	Primary checks
320 x 568	Two-column grid, menu fit, viewer controls, reflow, safe close
390 x 844	Modern mobile baseline, touch targets, lazy load behavior
768 x 1024	Tablet grid transition, landscape/portrait viewer
1024 x 768	Desktop navigation and 5-column region
1440 x 900	7-column native thumbnail grid, modal maximum size
200% / 400% zoom	Focus visibility, menu, modal, reflow, text overlap

Implementation Roadmap

Stage	Deliverables	Exit criteria
1. Foundation	Content schema, route templates, design tokens, representative fixtures, CI	One issue with 6 fixture pages renders semantically and responsively
2. Pipeline	All masters ingested; thumbnails/full variants; validation report; contact sheets	163 masters and all derivatives pass integrity checks
3. Experience	Homepage, three galleries, native dialog viewer, mobile menu, history/focus behavior	Functional and accessibility matrices pass on target browsers
4. Launch	SEO metadata, sitemaps, headers, analytics/RUM, budgets, deployment and rollback	Production smoke test, monitoring, statement, and owner handoff complete

Ownership

Role	Accountability
Editor / archivist	Issue titles, dates, synopses, rights, page descriptions/transcripts, final ordering
Designer	Responsive components, focus/contrast states, viewer behavior, approved image quality corpus
Engineer	Build, image pipeline, progressive enhancement, tests, deployment and monitoring
Accessibility reviewer	Keyboard and AT validation, alt-text policy, conformance evidence
Product owner	Scope decisions, acceptance sign-off, content and operational ownership

The critical path is content integrity, not interface complexity. Validate and lock the 163-page manifest early; the gallery, viewer, sitemap, and QA fixtures all depend on that single source of truth.

A

DATA CONTRACT

Appendix A: Sample Issue Manifest

```
{
  "id": "issue-01",
  "title": "Issue 01",
  "published": "YYYY-MM",
  "pageCount": 52,
  "coverPage": 1,
  "pages": [
    {
      "number": 17,
      "label": "Page 17",
      "description": "Optional concise editorial description",
      "master": "issue-01-page-017-master.tif",
      "width": 1224,
      "height": 864
    }
  ]
}
```

Schema rules

- id and URL slug are immutable, lowercase, and unique.
- pageCount equals pages.length and matches the issue contract.
- number is an integer between 1 and pageCount; the set is contiguous with no duplicates.
- width and height must be exactly 1224 and 864 for every master in this release.
- description is editorial content, not raw OCR. If absent, UI still supplies issue/page position.
- Generated derivative paths are not author-entered. The build computes them from issue, page, role, dimensions, format, and optional hash.

Derived record

The build can add a non-authoritative generated manifest containing byte sizes, checksums, MIME types, responsive candidates, and CDN URLs. This artifact powers integrity tests and diagnostics; it should never replace the reviewed editorial source manifest.

B

TRADEOFFS AND RATIONALE

Appendix B: Decision Log

Question	Chosen approach	Rejected / deferred
One route per page?	No, one canonical route per issue	163 thin routes add index noise unless unique textual metadata exists.
Client-render gallery?	No, pre-render complete HTML	Adds JS dependency and weakens resilience/crawlability for little benefit.
Virtualize thumbnails?	No	74 lightweight thumbnail nodes are manageable; virtualization complicates focus and find-in-page.
Custom lightbox library?	No dependency initially	Native dialog plus small controller covers requirements with lower weight and maintenance.
Force 153 px CSS width everywhere?	No; preserve asset dimension, allow narrow responsive rendering	A 320 px viewport cannot always fit two 153 px items plus padding/gap without overflow.
Loop previous/next at ends?	No; disable at first/last	Stopping preserves clear sequence boundaries; looping can disorient users.
Prefetch all full pages?	No	Wasteful. Prefetch only adjacent pages after intent and network checks.

R

AUTHORITATIVE SOURCES CONSULTED

References

- [1] W3C WAI - What's New in WCAG 2.2
<https://www.w3.org/WAI/standards-guidelines/wcag/new-in-22/>
- [2] W3C WAI - Understanding Focus Not Obscured (Minimum)
<https://www.w3.org/WAI/WCAG22/Understanding/focus-not-obscured-minimum>
- [3] W3C WAI - Understanding Target Size (Minimum)
<https://www.w3.org/WAI/WCAG22/Understanding/target-size-minimum>
- [4] W3C WAI-ARIA APG - Modal Dialog Pattern
<https://www.w3.org/WAI/ARIA/apg/patterns/dialog-modal/>
- [5] MDN - dialog HTML element
<https://developer.mozilla.org/en-US/docs/Web/HTML/Reference/Elements/dialog>
- [6] web.dev - Web Vitals
<https://web.dev/articles/vitals>
- [7] web.dev - Browser-level image lazy loading
<https://web.dev/articles/browser-level-image-lazy-loading>
- [8] Google Search Central - Image SEO best practices
<https://developers.google.com/search/docs/appearance/google-images>
- [9] MDN - Responsive images and img sizing
https://developer.mozilla.org/en-US/docs/Web/HTML/Guides/Responsive_images
- [10] Google Search Central - Learn about sitemaps
<https://developers.google.com/search/docs/crawling-indexing/sitemaps/overview>
- [11] Google Search Central - Canonical URL methods
<https://developers.google.com/search/docs/crawling-indexing/consolidate-duplicate-urls>
- [12] W3C - Web Content Accessibility Guidelines 2.2
<https://www.w3.org/TR/WCAG22/>

Research note: sources were reviewed on 13 July 2026. Standards and browser guidance evolve; verify compatibility data, search documentation, and organizational policy again before production launch. Recommendations in this report are implementation judgments derived from the cited standards and the stated 3-issue, 163-page inventory.